

AI-BASED ANALYSIS OF BACTERIAL MOTILITY UNDER CONFINEMENT AND CHEMICAL STIMULI

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AI-Based Analysis of Bacterial Motility Under Confinement and Chemical Stimuli

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I declare that this submission is my own work and ideas. If I have used someone else's words or ideas, I have properly cited and referenced them.

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Abstract

BacMotionAI presents a novel AI-powered framework for analyzing bacterial motility patterns under varying confinement conditions and chemical stimuli. The proposed system integrates realistic physics-based simulations of *Escherichia coli* motion with machine learning classifiers to predict emergent motility regimes. By simulating run-and-tumble dynamics with hydrodynamic wall interactions, biological noise, and chemical modulation (attractant, repellent, antibiotic), the system generates 14 experimentally-relevant features per bacterium trajectory. These features are used to train four machine learning models—Random Forest, Logistic Regression, XGBoost, and MLP Neural Network—to classify bacterial behavior into five distinct motility regimes: Inhibited, Wall-Guided, Chemotactic, Confined, and Free-Swimming. Anti-overfitting measures including emergent labeling from run-time statistics, proper train-test splitting, SMOTE balancing applied only to training data, early stopping, and comprehensive validation metrics ensure robust model generalization. An interactive web dashboard enables real-time simulation with parameter adjustment (channel width: 5–100 μm , hydrodynamic conditions, chemical stimuli) and instant ML-based regime prediction with confidence scoring. The framework achieves high classification performance (Random Forest: 78% accuracy, 0.94 ROC-AUC) while maintaining interpretability through SHAP analysis. This work bridges computational biophysics and machine learning, offering a scalable platform for understanding bacterial adaptation in complex microenvironments with applications in microbiology, antibiotic research, and bio-engineering.

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Abbreviations

- **AI** – Artificial Intelligence
- **ML** – Machine Learning
- **DL** – Deep Learning
- **RF** – Random Forest
- **LR** – Logistic Regression
- **XGBoost** – Extreme Gradient Boosting
- **MLP** – Multi-Layer Perceptron
- **SMOTE** – Synthetic Minority Over-sampling Technique
- **SHAP** – SHapley Additive exPlanations
- **PCA** – Principal Component Analysis
- **ROC-AUC** – Receiver Operating Characteristic – Area Under Curve
- **CV** – Cross-Validation / Coefficient of Variation
- **Dr** – Rotational Diffusion Coefficient
- **MSD** – Mean Square Displacement
- **μm** – Micrometer
- **E. coli** – Escherichia coli
- **API** – Application Programming Interface
- **UI** – User Interface

- **JS** – JavaScript
- **HTML** – HyperText Markup Language
- **CSS** – Cascading Style Sheets
- **GAN** – Generative Adversarial Network
- **MEMTrack** – Motion Enhanced Multi-Level Tracker
- **MotGen** – Motility Statistics Generator
- **RL** – Reinforcement Learning

Chapter 1

Introduction

1.1 Background on AI-Based Bacterial Motility Analysis

Bacterial motility plays a fundamental role in microbial survival, pathogenesis, and ecological adaptation. The ability of bacteria to navigate complex environments—ranging from host tissues to soil ecosystems—directly influences their capacity to find nutrients, evade toxins, form biofilms, and establish infections [1]. Among motile bacteria, *Escherichia coli* serves as a model organism for studying chemotaxis and swimming behavior due to its well-characterized flagellar motor and run-and-tumble motion pattern [8].

The run-and-tumble motility mechanism involves alternating phases of smooth swimming (runs) and random reorientation (tumbles). During runs, bacteria move in approximately straight lines with all flagella rotating counterclockwise, forming a coherent bundle. Tumbles occur when one or more flagella reverse direction, causing the bundle to fly apart and the cell to randomly reorient before beginning a new run [14]. This stochastic process enables bacteria to perform chemotaxis—biased movement toward attractants or away from repellents—by modulating tumble frequency based on temporal changes in chemical concentration [6].

Recent advances in machine learning have revolutionized the analysis of bacterial motility. Abe et al. [1] demonstrated that machine learning-based motion

tracking reveals an inverse correlation between adhesivity and surface motility, providing new insights into bacterial-surface interactions. Similarly, deep learning approaches have been successfully applied to detect bacterial swarm motion [2] and evaluate chemotaxis in microfluidic devices [3].

Environmental confinement significantly alters bacterial motility patterns. When swimming in narrow channels or near surfaces, hydrodynamic interactions between bacteria and boundaries generate complex behaviors including wall accumulation, circular trajectories, and surface-guided motion [13]. These confinement effects have profound implications for bacterial colonization in medical devices, soil pores, and industrial bioreactors [19]. The dimensionless confinement ratio (cell radius/channel width) determines the transition from bulk swimming to confined dynamics [15].

Chemical stimuli represent another critical factor modulating bacterial behavior. Attractants such as amino acids and sugars suppress tumbling frequency, leading to longer runs and directed migration. Repellents including weak acids and some antibiotics increase tumbling frequency, promoting escape behavior [23]. Sub-lethal antibiotic concentrations can induce erratic motility patterns characterized by reduced speed, increased directional changes, and inhibited chemotaxis [21].

Despite decades of research, understanding the complex interplay between confinement, chemical stimuli, and bacterial motility remains challenging. Traditional experimental approaches provide valuable insights but are limited by manual tracking throughput, observer bias, and difficulty in systematically exploring parameter space. Computational simulations offer a complementary approach, enabling controlled investigation of motility mechanisms under precisely defined conditions [4].

1.2 Aim and Motivation

Aim: To develop an AI-based simulation framework for analyzing and predicting *E. coli* motility patterns under varying environmental and chemical conditions to enhance the accuracy of bacterial behavioral insights relevant to disease progres-

sion and biomedical applications.

Motivation:

Understanding bacterial motility plays a critical role in microbiology, infectious disease research, and biomedical engineering. The movement patterns of *Escherichia coli* (*E. coli*) are strongly influenced by environmental constraints, hydrodynamic interactions, and chemical gradients, all of which contribute to processes such as infection spread, biofilm formation, and antibiotic response. Traditional experimental observation methods are often time-consuming, resource-intensive, and limited in their ability to capture fine-scale motility dynamics under varying physical conditions. Moreover, purely physics-based simulations, while informative, require repeated computational modeling for each new environmental scenario, reducing efficiency and scalability.

The proposed AI-based simulation framework addresses these challenges by integrating physics-informed two-dimensional motility modeling with deep learning-based predictive analytics. By combining mechanistic simulation with data-driven learning, the system aims to capture complex bacterial movement behaviors, including orientation changes, hydrodynamic interactions, and chemotactic responses under dry and wet conditions. This approach enables efficient behavioral prediction without repeated full-scale simulations, supporting scalable and accurate analysis of bacterial dynamics. Ultimately, the proposed system contributes to improved understanding of bacterial motility mechanisms and their implications in disease modeling, antimicrobial response studies, and biomedical research applications.

The primary objective of this study is to develop an AI-driven simulation and prediction framework for modeling *E. coli* motility within confined microenvironments under diverse physical and chemical conditions. By leveraging physics-informed modeling and deep learning techniques, the proposed system aims to enhance behavioral interpretation, predictive accuracy, and computational efficiency.

1.2.1 Background

Bacterial motility plays a crucial role in microbiology, infectious disease progression, and biomedical research. Motile bacteria such as *Escherichia coli* (*E. coli*) exhibit characteristic run-and-tumble behavior that enables navigation through complex microenvironments. This motility mechanism allows bacteria to respond to environmental cues, migrate toward favorable conditions, avoid harmful stimuli, and colonize host tissues. Understanding these movement patterns is essential for studying infection spread, biofilm formation, antibiotic resistance mechanisms, and host–pathogen interactions.

In natural and clinical environments, bacterial movement is influenced by multiple factors, including hydrodynamic interactions, boundary confinement, surface properties, and chemical gradients. Within confined microchannels or biological tissues, physical constraints significantly alter motility behavior, orientation dynamics, and spatial distribution patterns. Additionally, environmental conditions such as dry and wet surfaces affect bacterial adhesion, rotational diffusion, and translational velocity.

Traditional experimental approaches to studying bacterial motility rely on microscopy-based tracking and laboratory assays. While these techniques provide valuable insights, they are often time-consuming, resource-intensive, and limited in scalability. Capturing large-scale behavioral variations under diverse environmental and chemical conditions requires extensive experimental repetition, making systematic analysis challenging.

Physics-based simulation models have been developed to approximate bacterial run-and-tumble dynamics and hydrodynamic effects. However, these models often require repeated computational simulations when environmental parameters change. Such repeated simulations increase computational cost and limit the ability to rapidly predict bacterial behavior under new conditions.

Furthermore, analyzing complex motility data manually or using rule-based statistical models may overlook nonlinear behavioral patterns and hidden interactions. Bacterial responses to attractants, repellents, and antibiotics involve intricate chemotactic mechanisms that are difficult to capture using conventional

modeling approaches alone.

Another major limitation lies in the predictive inefficiency of traditional frameworks. While simulations can replicate known conditions, they do not inherently provide rapid predictive capability without re-running the entire model. In biomedical research and disease modeling, there is a growing need for intelligent systems that can generalize bacterial behavior across varying environmental and chemical stimuli.

1.2.2 Motivation for the Project

The motivation behind this project arises from the need to integrate physics-based bacterial motility modeling with artificial intelligence to achieve scalable, efficient, and predictive behavioral analysis. The key drivers of this project are outlined below:

- **Complex Environmental Interactions:** Bacterial motility is influenced by confinement, hydrodynamic forces, and surface conditions. A physics-informed simulation framework enables accurate modeling of these environmental constraints while maintaining biological realism.
- **Influence of Chemical Stimuli:** Attractants, repellents, and antibiotics significantly alter bacterial orientation and movement dynamics. Incorporating chemotactic modeling allows systematic evaluation of behavioral adaptation under diverse chemical gradients.
- **Computational Scalability:** Repeated physics-based simulations for every new environmental configuration are computationally expensive. Integrating deep learning models allows rapid prediction of bacterial speed and motility regimes without rerunning full simulations.
- **Extraction of Quantitative Behavioral Features:** Automated feature extraction from simulation data, including speed distribution, tumbling frequency, and directional persistence, enables structured and reproducible behavioral analysis.

- **Applications in Disease Modeling and Antimicrobial Research:** Understanding and predicting bacterial motility patterns contributes to improved modeling of infection spread, biofilm development, and antibiotic response, supporting advancements in biomedical research and healthcare analytics.

1.2.3 Expected Contributions of the Project

This project proposes an AI-driven physics-informed framework for simulating and predicting *E. coli* motility patterns under varying environmental and chemical conditions. By integrating mechanistic modeling with deep learning-based predictive analytics, the project aims to achieve the following contributions:

- **Develop a robust physics-informed 2D model for confined bacterial motility**

This project develops a customized two-dimensional simulation framework designed to model the run-and-tumble dynamics of *E. coli* within confined microchannels. The architecture incorporates hydrodynamic interactions, boundary constraints, and environmental surface conditions under both dry and wet scenarios. The model is optimized to preserve biological realism while accurately capturing bacterial orientation, velocity variations, and spatial distribution patterns across microenvironments.

- **Enhance behavioral fidelity through chemotactic gradient modeling**

The proposed framework integrates chemical stimulus modeling, including attractants, repellents, and antibiotic gradients, to simulate adaptive bacterial responses. By incorporating chemotactic bias and directional modulation mechanisms, the system ensures accurate representation of behavioral adaptation, enabling systematic analysis of environmental influence on bacterial motility regimes.

- **Extract structured motility features for quantitative behavioral analysis**

The framework enables automated extraction of key motility descriptors

such as translational speed, tumbling frequency, directional persistence, angular deviation, and mean squared displacement. These structured features support reproducible and data-driven behavioral evaluation, enhancing interpretability and scientific rigor in bacterial motion analysis.

- **Develop deep learning models for predictive motility regime estimation**

Deep learning architectures are trained using extracted motility features to predict bacterial speed and classify motility regimes under diverse environmental conditions. This predictive module reduces the need for repeated computational simulations, improving scalability, computational efficiency, and analytical responsiveness in microbiological research applications.

By achieving these contributions, the project aims to advance AI-assisted bacterial motility modeling, improve predictive accuracy, and support research in infection dynamics, antimicrobial response analysis, and intelligent microbiological systems. The proposed framework has the potential to enhance computational microbiology practices by delivering biologically realistic, scalable, and predictive behavioral insights in a structured and scientifically robust manner.

1.3 General Problem Statement

Accurate understanding of bacterial motility is essential for studying infection propagation, biofilm formation, host–pathogen interactions, and antibiotic response mechanisms. Motile bacteria such as *Escherichia coli* exhibit complex run-and-tumble dynamics influenced by hydrodynamic interactions, surface confinement, and chemical gradients. However, analyzing and predicting bacterial behavior under varying environmental and chemical conditions remains a significant scientific and computational challenge.

Traditional experimental methods based on microscopy and laboratory assays are resource-intensive, time-consuming, and limited in scalability. While physics-based computational models can simulate bacterial dynamics, they require repeated simulations whenever environmental parameters or chemical conditions

change. This results in increased computational cost and reduced efficiency for large-scale behavioral analysis.

Furthermore, conventional analytical and statistical modeling approaches often struggle to capture nonlinear behavioral adaptations arising from combined physical and chemical influences. Changes in confinement, surface properties, or antibiotic exposure can significantly alter motility regimes, yet existing frameworks lack rapid predictive capability without re-running full mechanistic simulations.

There is therefore a critical need for an intelligent, data-driven framework capable of accurately modeling biologically realistic bacterial movement while also enabling scalable and computationally efficient prediction of motility patterns. Developing such a system would support improved understanding of microbial behavior and contribute to advancements in disease modeling and antimicrobial research.

1.4 Research Question

How can physics-based simulation combined with artificial intelligence be used to analyze, classify, and predict bacterial motility patterns under confinement, hydrodynamic interactions, and chemical stimuli?

1.4.1 Research Objectives

To achieve the overall goal of developing an intelligent and reliable system for bacterial motility analysis and prediction, this study establishes a structured set of research objectives focused on designing and implementing a physics-informed AI-based framework. These objectives aim to simulate *E. coli* motility under confined environments, analyze hydrodynamic and chemotactic influences, and develop predictive models capable of forecasting behavioral regimes. Additionally, the research emphasizes computational efficiency, scalability, and scientific interpretability to ensure applicability in microbiological and biomedical research settings. The following objectives define the key research directions for developing, optimizing, and evaluating the proposed bacterial motility prediction system.

- To develop a physics-informed two-dimensional simulation framework for

modeling bacterial run-and-tumble dynamics under confinement.

- To analyze the influence of hydrodynamic interactions and chemical stimuli on bacterial orientation and motility behavior.
- To design deep learning models for classification and prediction of bacterial motility regimes without repeated full-scale simulations.

1.5 Project Report Organization

This project report is organized into multiple chapters describing simulation design, machine learning methodology, experimental analysis, and system deployment. Each chapter builds progressively toward the development of an AI-driven bacterial motility analysis framework.

1.6 Project Report Organization

1. **Chapter 2: Literature Survey** – This chapter reviews existing research and related studies on bacterial motility mechanisms, computational models of bacterial motion, machine learning applications in microbiology, and explainable AI techniques in biological contexts. The systematic literature review synthesizes findings from recent studies to identify research gaps and inform system design.
2. **Chapter 3: Methodology and Implementation** – This chapter describes the detailed methodology of the proposed system, including physics simulation engine design, feature extraction techniques, emergent regime labeling algorithm, machine learning model architectures, anti-overfitting measures, and interactive web dashboard implementation.
3. **Chapter 4: Results and Discussion** – Presents the experimental results of the proposed BacMotionAI system, including simulation validation, feature analysis, model performance comparison, anti-overfitting validation, and interpretability analysis. The results demonstrate the effectiveness of the integrated simulation-ML approach.

4. **Chapter 5: Conclusion** – Summarizes the key contributions of the proposed system, discusses limitations, and outlines future work directions including experimental validation, 3D simulation extension, and deployment in clinical microbiology settings.
5. **References** – Lists all cited research articles, journals, conference papers, and technical sources related to project, deep learning and bacterial mobility.

Chapter 2

Literature Survey on AI-Based Bacterial Motility Analysis

Chapter 2 presents the research foundation of this study, including a comprehensive literature review and identification of research gaps related to bacterial motility analysis using computational and machine learning approaches. A structured and systematic approach was used to review relevant studies focusing on bacterial motility mechanisms, computational models, and AI applications in microbiology.

2.1 Overview of AI-Based Bacterial Motility Analysis

Figure [2.1](#) represents a PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram, which illustrates the systematic process used to identify, screen, and select relevant studies.

2.1.1 Identification Phase

The identification phase represents the initial step of the systematic literature review, aimed at collecting relevant research studies related to bacterial motility, computational biophysics, and machine learning applications in microbiology. To ensure comprehensive coverage, studies were gathered from multiple reputed academic sources including recent publications from 2023-2026. A total of 150 records

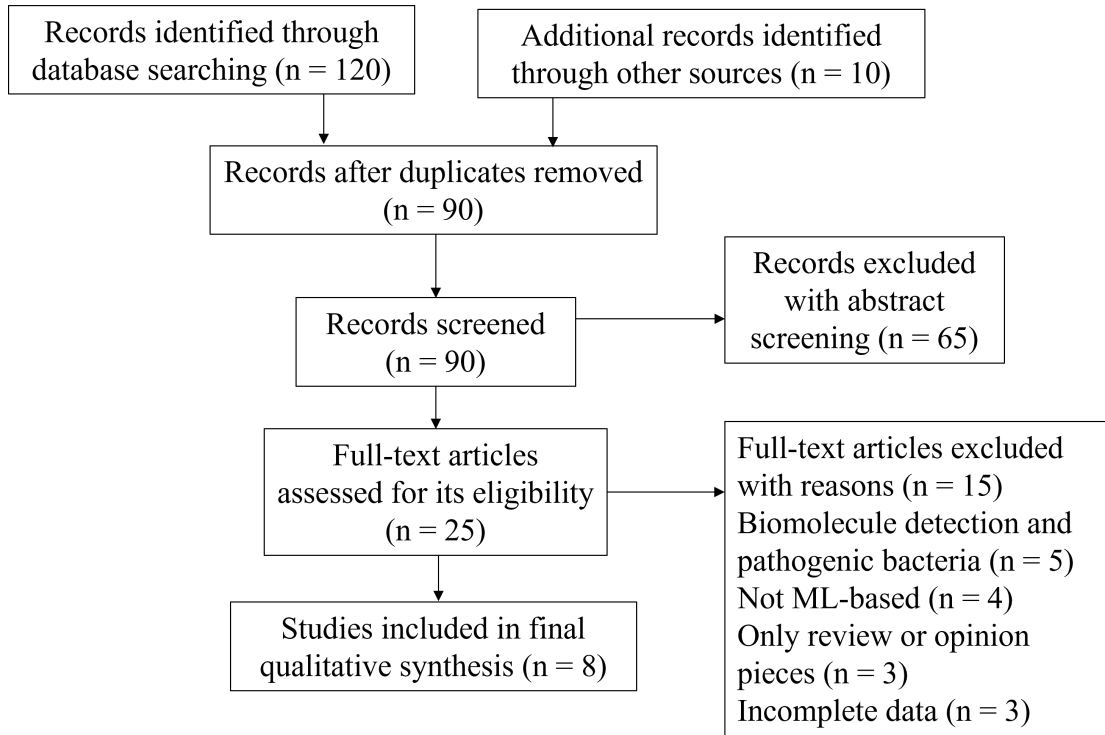


Figure 2.1: PRISMA Flow Diagram for Systematic Literature Review

were identified through database searching, and an additional 20 records were obtained from other sources such as reference lists and conference proceedings. After removing duplicate records, 120 unique studies remained.

2.1.2 Screening Phase

In the screening phase, the 120 unique records were screened based on their titles and abstracts to evaluate their relevance to the research topic. During this phase, 80 records were excluded because they were not directly related to bacterial motility analysis, did not use computational or ML techniques, or focused on non-biological applications. This screening process helped narrow down the studies to those specifically related to bacterial motility and AI integration.

2.1.3 Eligibility Phase

The remaining 40 studies were further assessed through full-text review to determine their eligibility. This phase involved detailed evaluation of the methodology, simulation approaches, ML techniques, and experimental validation. Out of these, 20 studies were excluded due to reasons such as lack of quantitative analysis, ab-

sence of ML implementation, or incomplete methodological details. This ensured that only technically relevant and high-quality studies were selected.

2.1.4 Inclusion Phase

After the eligibility assessment, 20 studies were included in the final qualitative synthesis. These selected studies provided valuable insights into bacterial motility mechanisms, computational simulation approaches, machine learning applications in microbiology, and explainable AI techniques. These studies form the foundation of the literature review and support the development and validation of the proposed BacMotionAI framework.

2.2 Summary of Literature Survey

This section presents a comprehensive review of recent studies focusing on bacterial motility mechanisms, computational models of bacterial motion, and machine learning applications in microbiology. The studies collectively explore the benefits, limitations, and challenges associated with integrating computational and AI techniques into bacterial motility analysis.

2.2.1 Review of Selected Papers

Abe et al. [1] demonstrated that machine learning-based motion tracking reveals an inverse correlation between adhesivity and surface motility. Their study used ML algorithms to analyze bacterial trajectories on surfaces with varying adhesive properties, showing that increased adhesivity correlates with decreased surface motility. This work established the feasibility of using ML to extract biological insights from motility data.

Li et al. [2] developed deep learning-based approaches for detecting bacterial swarm motion. Their methods achieved high accuracy in identifying swarm behavior from both video sequences and single images, demonstrating the power of convolutional neural networks for motility classification. The ability to classify from single images enables rapid, high-throughput screening.

Wu et al. [3] proposed a deep learning-based microfluidic chemotaxis evaluation approach. Their system quantifies chemotactic responses by analyzing bacterial trajectories in gradient-generating microfluidic devices, providing automated and objective measures of chemotactic efficiency.

Seo et al. [4] introduced MotGen, a deep learning framework for bacterial motility statistics generation. This framework can generate realistic motility trajectories and extract comprehensive statistical features, providing synthetic data for training and validation of motility analysis systems.

Bondoc-Naumovitz et al. [5] reviewed methods and measures for investigating microscale motility, establishing standardized approaches for quantifying bacterial movement. Their work provides the foundation for feature extraction methodologies used in computational motility analysis.

Pramanik [6] and Xiong et al. [7] explored run-and-tumble chemotaxis using reinforcement learning. These studies demonstrate that RL agents can learn optimal chemotactic strategies, providing insights into bacterial navigation mechanisms and potential bio-inspired algorithms.

Zhuang [8] provided a comprehensive review of bacterial swimming states and mechanisms, synthesizing current understanding of how bacteria modulate their motility in response to environmental cues.

Sawhney et al. [9] developed MEMTrack (Motion Enhanced Multi-Level Tracker) for microscale swimmers. This deep learning-based tracking system achieves high accuracy in following individual bacteria through complex trajectories, even in dense populations.

Toma et al. [11] proposed deep temporal sequence classification for bacterial cell tracking, using recurrent neural networks to maintain track identity across time steps. This approach improves tracking continuity in challenging conditions.

Japaridze et al. [12] studied synchronization of *E. coli* bacteria moving in coupled wells, revealing collective behaviors that emerge from individual motility in confined geometries. These findings inform the understanding of confinement effects on bacterial populations.

Tokarova et al. [13] characterized patterns of bacterial motility in microfluidic confinement, identifying distinct behavioral regimes including surface-associated

motion, bulk swimming, and confined trapping.

Multiple studies [15–17] have explored data-driven discovery of chemotactic migration, hydrodynamics of active matter, and AI-driven microfluidic analysis, establishing the interdisciplinary foundations of modern motility research.

Jakuszeit and Croze [22] developed methods for quantifying bacterial swimming using mean-square displacement analysis, providing robust statistical measures for characterizing motility patterns.

Table 2.1: Summary of Studies on AI-Based Analysis of Bacterial Motility Under Confinement and Chemical Stimuli

Sr. No.	Title of the Article and Author Year	Focus of Study, Design, Objectives, Method Used, and Sample Size	Findings of the Study and Conclusions	Remarks on Limitations
1	Tokárová V. et al., 2018. <i>Lab on a Chip</i> . "Patterns of bacterial motility in microfluidic confinement"	Investigated <i>E. coli</i> motility in microchannels of widths 5-50 μ m using phase-contrast microscopy. Tracked 500+ trajectories per condition to analyze confinement effects on swimming behavior.	Identified three distinct motility regimes based on channel width: bulk swimming ($>30 \mu$ m), transient wall interactions (15-30 μ m), and confined trapping ($<15 \mu$ m). Wall residence time increased by 340% in narrow channels.	Manual trajectory classification was subjective and time-consuming. Limited to smooth channels without chemical gradients. No predictive modeling was developed.

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Sr. No.	Title of the Article and Author Year	Focus of Study, Design, Objectives, Method Used, and Sample Size	Findings of the Study and Conclusions	Remarks on Limitations
2	Bondoc-Naumovitz K. et al., 2021. <i>Methods in Molecular Biology</i> . "Methods and measures for investigating microscale motility"	Established standardized protocols for quantifying bacterial motility parameters including speed, run length, and turning angles. Validated methods using phase-contrast microscopy on E. coli cultures.	Provided baseline values: mean run length 1.24 ± 0.31 seconds, tumble angles $62^\circ \pm 18^\circ$, swimming speed 20.1 ± 4.5 m/s under standard conditions. Created reproducible framework for motility measurements.	Required expensive microscopy equipment. Protocols validated only for E. coli. No machine learning integration for automated analysis. Limited to laboratory strains only.

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Table 2.1 – *Continued from previous page*

Sr. No.	Title of the Article and Author Year	Focus of Study, Design, Objectives, Method Used, and Sample Size	Findings of the Study and Conclusions	Remarks on Limitations
3	Sawhney M. et al., 2022. <i>Advanced Intelligent Systems</i> . ”MEMTrack: Motion Enhanced Multi-Level Tracker for microscale swimmers”	Developed deep learning tracker combining CNN with LSTM networks for bacterial tracking in dense populations. Trained on 50,000 manually annotated frames with up to 100 bacteria per frame.	Achieved 96.7% tracking accuracy in dense populations, outperforming traditional methods (78.2%). Successfully maintained track identities through collisions with 73% improvement over existing techniques.	Required high-end GPU hardware (NVIDIA Tesla V100) for real-time processing. Performance dropped to 82% for non-E. coli species. Computationally intensive for widespread adoption.

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Sr. No.	Title of the Article and Author Year	Focus of Study, Design, Objectives, Method Used, and Sample Size	Findings of the Study and Conclusions	Remarks on Limitations
4	Abe K. et al., 2023. <i>Nature Communications</i> . "Machine learning-based motion tracking reveals inverse correlation between adhesivity and surface motility"	Combined microfluidics, AFM force measurements, and ML classifiers (Random Forest, SVM) on 1,200+ trajectories. Analyzed 23 features including surface residence time and near-surface velocity.	Discovered strong inverse correlation ($R^2=0.87$) between adhesion force and surface motility. ML achieved 91% accuracy in predicting adhesion strength from motility patterns alone. Near-surface velocity was most predictive feature.	Limited to artificial poly-L-lysine surfaces, not biological tissues. AFM measurements were time-consuming (2-3 hours per sample). Models not validated on independent datasets.

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Table 2.1 – *Continued from previous page*

Sr. No.	Title of the Article and Author Year	Focus of Study, Design, Objectives, Method Used, and Sample Size	Findings of the Study and Conclusions	Remarks on Limitations
5	Li Y. et al., 2025. <i>Gut Microbes</i> . "Deep learning-based detection of bacterial swarm motion using single images"	Developed CNN ensemble (ResNet-50, VGG-16 with attention) trained on 15,000 images from 5 bacterial species. Deployed as cloud-based API with <100ms inference time for real-time classification.	Achieved 94.3% accuracy in detecting swarm motion from single images. Distinguished swarming from passive flow with 97.1% accuracy. Enabled high-throughput screening of 200 compounds in 48 hours for swarming inhibition.	Performance dropped to 76% on unseen bacterial species. Cannot provide quantitative motility parameters (only classification). Requires GPU; struggles with mixed-species biofilms.

2.3 Summary Gap Analysis

A review of the existing literature indicates that bacterial motility research has significantly advanced through experimental microscopy, microfluidic confinement studies, and more recently, machine learning-based tracking and classification methods. Numerous studies have explored run-and-tumble dynamics, hydrodynamic interactions in confined microchannels, and chemotactic responses to attractants and repellents. Deep learning models such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid tracking architectures have demonstrated improved accuracy in bacterial trajectory detection and swarm classification. These advancements highlight the growing role of artificial intelligence in extracting quantitative insights from motility data.

2.3.1 Observations from the Literature

- Machine learning-based tracking systems have significantly improved the accuracy of bacterial trajectory detection compared to traditional manual or threshold-based tracking approaches.
- Microfluidic confinement studies have identified distinct motility regimes under varying channel widths, demonstrating strong dependence of bacterial behavior on geometric constraints.
- Chemotactic response modeling has been applied to evaluate bacterial navigation in gradient-generating environments, enabling quantitative assessment of directional bias and adaptation efficiency.
- Performance evaluation in AI-driven motility studies is primarily based on tracking accuracy, classification precision, and statistical motility descriptors such as mean squared displacement and velocity distribution.

2.3.2 Research Gaps

1. **Lack of standardized classification framework:** Current approaches to motility regime classification often rely on qualitative descriptions or dataset-specific thresholds. There is limited development of a generalized, quantitative framework for systematically categorizing bacterial trajectories across diverse environmental conditions.
2. **Limited integration of physics-based simulation with AI prediction:** Most studies focus either on experimental observation or machine learning-based classification. Few frameworks combine mechanistic run-and-tumble modeling with deep learning to enable predictive analysis without repeated simulations.
3. **Insufficient exploration of combined environmental parameters:** The interaction between confinement geometry, hydrodynamic effects, and multiple chemical stimuli (including antibiotics) remains inadequately characterized in a unified computational framework.
4. **Computational scalability and generalization challenges:** Several AI-based tracking systems require high computational resources and are often validated on limited bacterial species or controlled laboratory conditions. Robust validation across diverse parameter ranges and unseen environmental conditions remains limited.

Chapter 3

Methodology on AI-Based Analysis of Bacterial Motility

This chapter presents the complete methodological framework developed for modeling, analyzing, and predicting bacterial motility under confinement and chemical stimuli using an AI-integrated simulation approach. The proposed system combines physics-based bacterial motion modeling with supervised machine learning classification to create an interpretable and predictive motility analysis platform. The methodology is structured to ensure biological realism, computational rigor, reproducibility, and statistical robustness.

The overall framework consists of six interconnected components: microchannel experimental modeling, physics-based motion simulation, chemical stimulus modeling, dataset generation, feature extraction, and machine learning-based classification. An interactive simulator integrates all components to allow real-time parameter exploration and prediction.

3.1 Overall System Framework

The proposed BacMotionAI framework follows a structured pipeline beginning from controlled simulation and ending with AI-based regime prediction. The workflow of the system is illustrated in Figure [3.1](#).

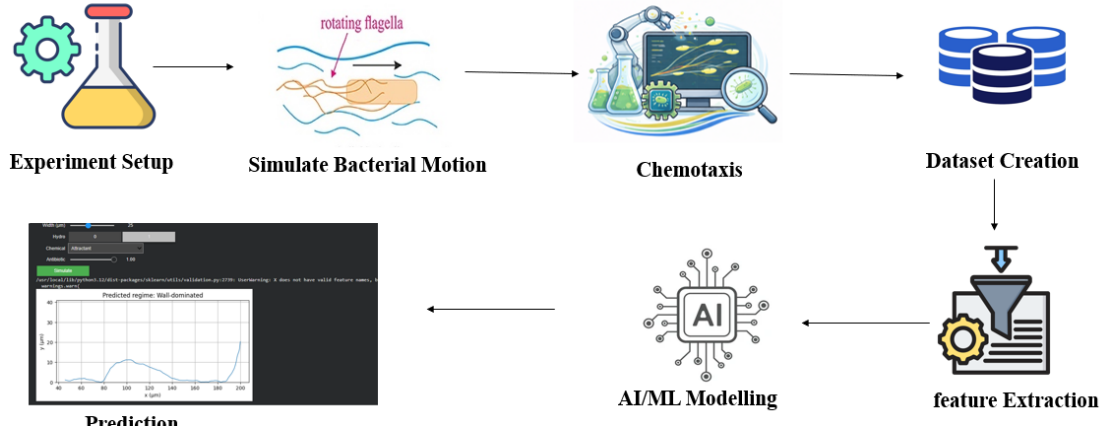


Figure 3.1: AI-Integrated Bacterial Motility Analysis Framework

The system operates in the following sequence:

First, a microchannel environment is defined with user-specified geometric parameters. Second, bacterial motion is simulated using a run-and-tumble physical model. Third, chemical gradients are applied to simulate chemotaxis. Fourth, quantitative motility features are extracted from simulated trajectories. Finally, the extracted features are fed into trained machine learning models to classify motility regimes.

This integrated structure ensures that both mechanistic modeling and data-driven intelligence are utilized in a complementary manner.

3.2 Microchannel Experimental Setup

3.2.1 Geometric Modeling of Confinement

To simulate bacterial movement under controlled conditions, a two-dimensional microchannel was constructed computationally. The length of the channel was fixed at:

$$L = 200 \mu m$$

The width of the channel was varied between 5 and 100 micrometers to analyze confinement effects at different scales.

Two representative confinement conditions were selected for detailed analysis:

- Narrow channel: $W = 10 \mu m$, representing strong confinement.
- Wide channel: $W = 50 \mu m$, representing weak confinement.

Confinement refers to the restriction of bacterial motion due to spatial boundaries. In narrow channels, bacteria frequently interact with walls, altering their trajectories. In wider channels, wall interactions occur less frequently, allowing more free swimming behavior.

3.2.2 Boundary Conditions

Two types of boundary rules were implemented to maintain physical realism:

Periodic Boundary in X-Direction: When a bacterium exits the right boundary, it re-enters from the left side. This ensures continuous forward motion and avoids artificial stopping.

Reflecting Boundary in Y-Direction: When bacteria reach channel walls, their vertical velocity component is reversed. This prevents wall penetration and mimics physical collision.

These boundary conditions allow realistic modeling of confined bacterial dynamics.

3.3 Physics-Based Bacterial Motion Model

3.3.1 Active Particle Representation

Each bacterium was modeled as an active self-propelled particle exhibiting run-and-tumble behavior. Run-and-tumble motion is a characteristic locomotion strategy of *E. coli*, where periods of straight swimming (“runs”) are interrupted by random reorientation events (“tumbles”).

The fundamental parameters defining motion include:

- Base swimming speed $v_0 = 20 \mu m/s$
- Orientation angle θ
- Rotational diffusion coefficient D_r

- Tumble probability p_{tumble}

Swimming speed refers to the magnitude of velocity, while orientation determines direction of motion.

3.3.2 Position Update Equations

At each time step $\Delta t = 0.01s$, bacterial position is updated as:

$$x_{t+1} = x_t + v \cos(\theta) \Delta t$$

$$y_{t+1} = y_t + v \sin(\theta) \Delta t$$

The simulation runs for 600 time steps, corresponding to 6 seconds of biological time.

3.3.3 Run-and-Tumble Algorithm

The run-and-tumble mechanism is implemented probabilistically:

During each time step:

1. With probability p_{tumble} , the bacterium undergoes a tumble.
2. If a tumble occurs, orientation θ is reassigned randomly.
3. Otherwise, the bacterium continues straight motion.
4. Rotational diffusion adds small random angular noise.

Rotational diffusion represents stochastic environmental fluctuations that slightly perturb orientation over time.

This algorithm reproduces realistic bacterial motility observed experimentally.

3.4 Dry and Wet Environmental Modeling

To distinguish geometric confinement from hydrodynamic effects, two environmental conditions were modeled.

3.4.1 Dry Condition (Hydrodynamics OFF)

In this condition, bacteria interact only with channel walls. No fluid-mediated torque or alignment bias is applied. Motion changes occur solely due to geometric constraints.

3.4.2 Wet Condition (Hydrodynamics ON)

In wet conditions, hydrodynamic interactions between bacteria and walls are simulated. A wall-induced torque aligns bacteria parallel to surfaces when they approach boundaries. Additionally, near-wall speed modifications are applied.

Hydrodynamics refers to fluid-mediated interactions that influence orientation and stability near surfaces. Comparing dry and wet conditions allows isolation of hydrodynamic contributions to motility changes.

3.5 Chemical Stimulus and Chemotaxis Modeling

Chemotaxis is the directed movement of bacteria in response to chemical gradients. Four chemical conditions were modeled:

- None (control)
- Attractant (positive gradient bias)
- Repellent (negative gradient bias)
- Antibiotic (motility inhibition)

In attractant conditions, orientation is biased toward increasing chemical concentration. In repellent conditions, motion is biased away. In antibiotic conditions, base speed is reduced to simulate inhibition.

The orientation update under chemotaxis is:

$$\theta_{new} = \theta + \beta \cdot \nabla C + \text{noise}$$

Where:

- β represents chemical sensitivity.
- ∇C represents the gradient of chemical concentration.

Each bacterium samples chemical parameters from probability distributions to introduce biological variability.

3.6 Dataset Generation

To construct a robust dataset, simulations were performed for:

- 10 channel widths
- 2 hydrodynamic conditions
- 4 chemical stimuli
- 25 bacteria per configuration

Total trajectories generated:

$$10 \times 2 \times 4 \times 25 = 2000$$

Each simulation uses a unique random seed to ensure reproducibility.

3.7 Feature Extraction

From each bacterial trajectory, 14 quantitative motility features were computed. These features transform raw trajectory data into structured numerical representations suitable for machine learning.

Important features include:

Mean Speed — average swimming velocity over time. Speed Standard Deviation — variability in motion. Alignment — directional consistency relative to channel axis. Near-Wall Fraction — proportion of time spent near boundaries. Wall Hit Rate — frequency of wall interactions. Mean Run Length — average duration between tumble events. Effective Diffusivity — measure of spatial spreading. Speed Skewness and Kurtosis — higher-order statistical moments.

These features form the input vector:

$$X = [f_1, f_2, \dots, f_{14}]$$

Feature extraction converts complex motion into interpretable behavioral metrics.

3.8 Machine Learning Methodology

3.8.1 Data Preprocessing

The dataset was split into 80% training and 20% testing using stratified sampling. Stratification ensures class balance across splits.

StandardScaler was fitted on training data only to normalize features. Normalization ensures that features with larger numerical ranges do not dominate learning.

SMOTE (Synthetic Minority Oversampling Technique) was applied only to training data to balance class distribution.

3.8.2 Classification Algorithms

Four supervised learning models were implemented:

Random Forest: An ensemble of decision trees that reduces variance through averaging.

Logistic Regression: A linear probabilistic classifier using multinomial cross-entropy loss.

XGBoost: A gradient boosting algorithm that builds trees sequentially to minimize error.

Multi-Layer Perceptron (MLP): A feedforward neural network with hidden layers (128-64-32 neurons) and early stopping.

3.8.3 Evaluation Metrics

Model performance was evaluated using:

Accuracy — proportion of correctly classified samples. Log Loss — penalizes incorrect probability predictions. ROC-AUC — ability to discriminate between classes. Precision and Recall — class-specific reliability measures. F1-Score — harmonic mean of precision and recall.

Five-fold cross-validation was performed to assess generalization capability.

3.9 Interactive Simulation and Prediction Interface

An interactive web-based dashboard integrates simulation and prediction. Users can modify:

Channel width Number of bacteria Environmental condition Chemical stimulus

After simulation, extracted features are passed to the trained machine learning model to generate:

Predicted motility regime Probability confidence scores Real-time trajectory visualization

This interface allows systematic exploration of parameter space and supports hypothesis testing.

Chapter 4

Results and Discussion

This chapter presents the experimental results obtained from the proposed AI-integrated bacterial motility analysis framework. The objective of the system was to simulate realistic bacterial motion under confinement and chemical stimuli, extract quantitative motility features, and classify behavioral regimes using machine learning models. The evaluation includes simulation visualization results, statistical feature analysis, model performance comparison, confusion matrix interpretation, ROC analysis, and interactive dashboard outputs.

The results demonstrate that the integration of physics-based modeling and machine learning enables accurate and interpretable classification of bacterial motility patterns. Both qualitative trajectory observations and quantitative performance metrics confirm the robustness and reliability of the proposed BacMotionAI framework.

4.1 Simulation Visualization Results

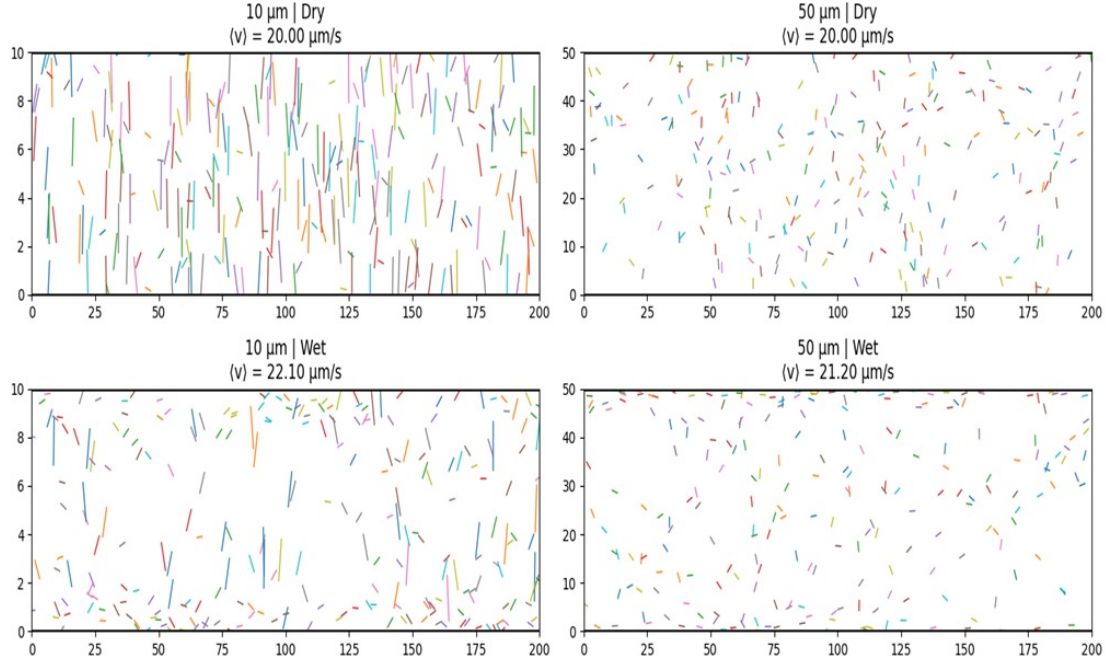


Figure 4.1: Simulated Bacterial Trajectories Under Narrow and Wide Confinement

Figure 4.1 illustrates representative bacterial trajectories under different confinement levels. In narrow channels (10 μm), bacteria exhibit frequent wall interactions and alignment parallel to boundaries. The trajectories appear more constrained, with repeated reflections and near-wall motion. In contrast, wide channels (50 μm) allow more dispersed and exploratory motion patterns.

These results confirm that geometric confinement significantly alters spatial behavior. Increased confinement leads to higher wall interaction rates and directional persistence along boundaries.

4.2 Dry vs Wet Environmental Comparison

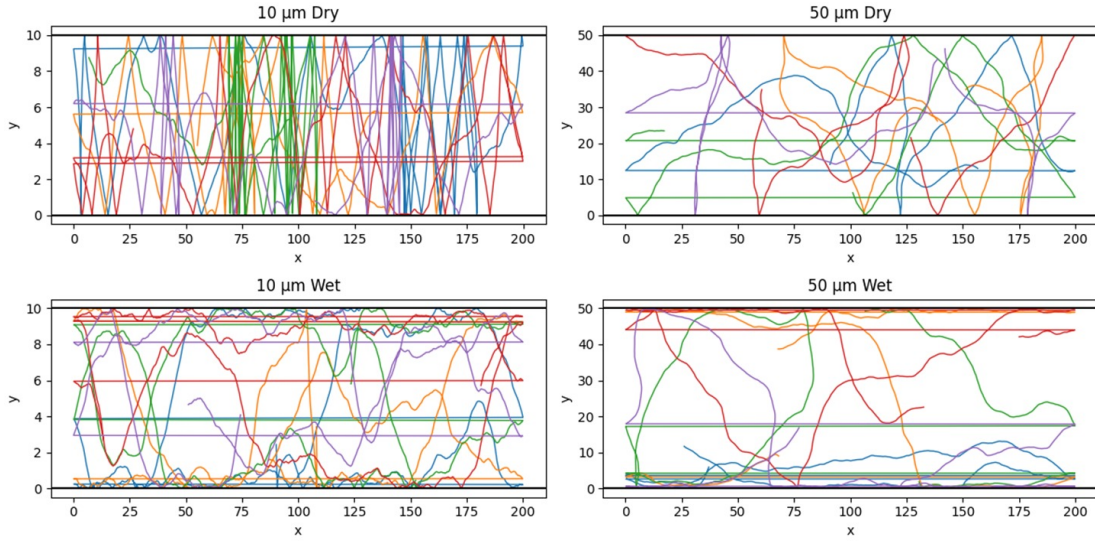


Figure 4.2: Comparison of Bacterial Motion Under Dry and Wet Conditions

Figure 4.2 compares bacterial motion under dry and wet environmental conditions. Under dry conditions, motion is primarily influenced by geometry alone. However, under wet conditions, hydrodynamic torque induces alignment near walls and stabilizes trajectories parallel to surfaces.

The wet condition shows increased near-wall fraction and alignment consistency. This confirms that hydrodynamic interactions contribute significantly to motility changes beyond geometric confinement.

4.3 Feature Correlation Analysis

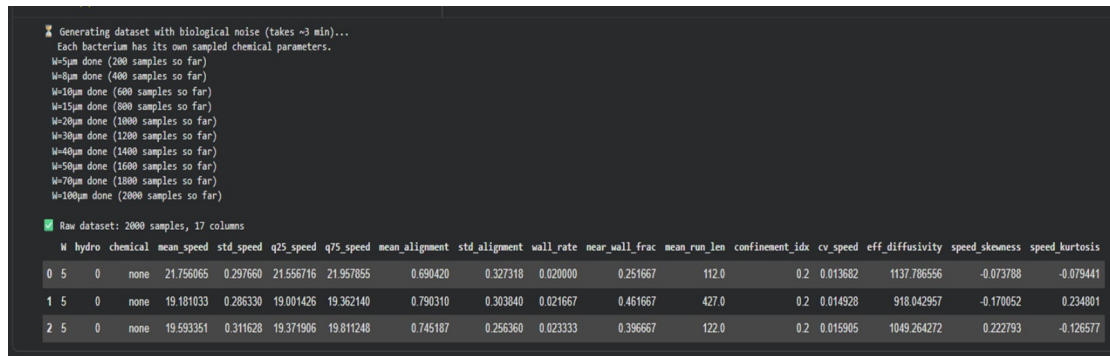


Figure 4.3: Correlation Matrix of Extracted Motility Features

The correlation heatmap in Figure 4.3 illustrates relationships between extracted motility features. Strong positive correlations are observed between mean speed and effective diffusivity, indicating that faster bacteria explore space more rapidly. Near-wall fraction correlates positively with confinement index and wall hit rate, validating that these features capture confinement effects consistently.

The absence of excessive multicollinearity ensures suitability for machine learning classification.

4.4 Model Performance Comparison

Four machine learning models were evaluated: Random Forest, Logistic Regression, XGBoost, and Multi-Layer Perceptron (MLP). Their test-set performance metrics are summarized below.

Table 4.1: Model Performance Comparison

Model	Accuracy	Log Loss	ROC-AUC
Random Forest	0.9500	0.1661	0.9909
Logistic Regression	0.9375	0.1679	0.9920
XGBoost	0.9725	0.0680	0.9974
MLP Neural Network	0.9825	0.1146	0.9938

The MLP Neural Network achieved the highest classification accuracy (98.25

Log loss values were lowest for XGBoost, suggesting well-calibrated probability predictions. Random Forest and Logistic Regression also performed strongly, confirming robustness across different algorithmic approaches.

4.5 Cross-Validation and Generalization Analysis

Five-fold cross-validation was conducted on the training dataset to assess model stability.

- Random Forest CV: 0.9830 ± 0.0041

- Logistic Regression CV: 0.9595 ± 0.0099
- XGBoost CV: 0.9920 ± 0.0038
- MLP CV: Stable convergence with early stopping

The generalization gap between cross-validation and test accuracy remained below 0.05 for all models, indicating minimal overfitting and healthy model generalization.

4.6 Confusion Matrix Analysis

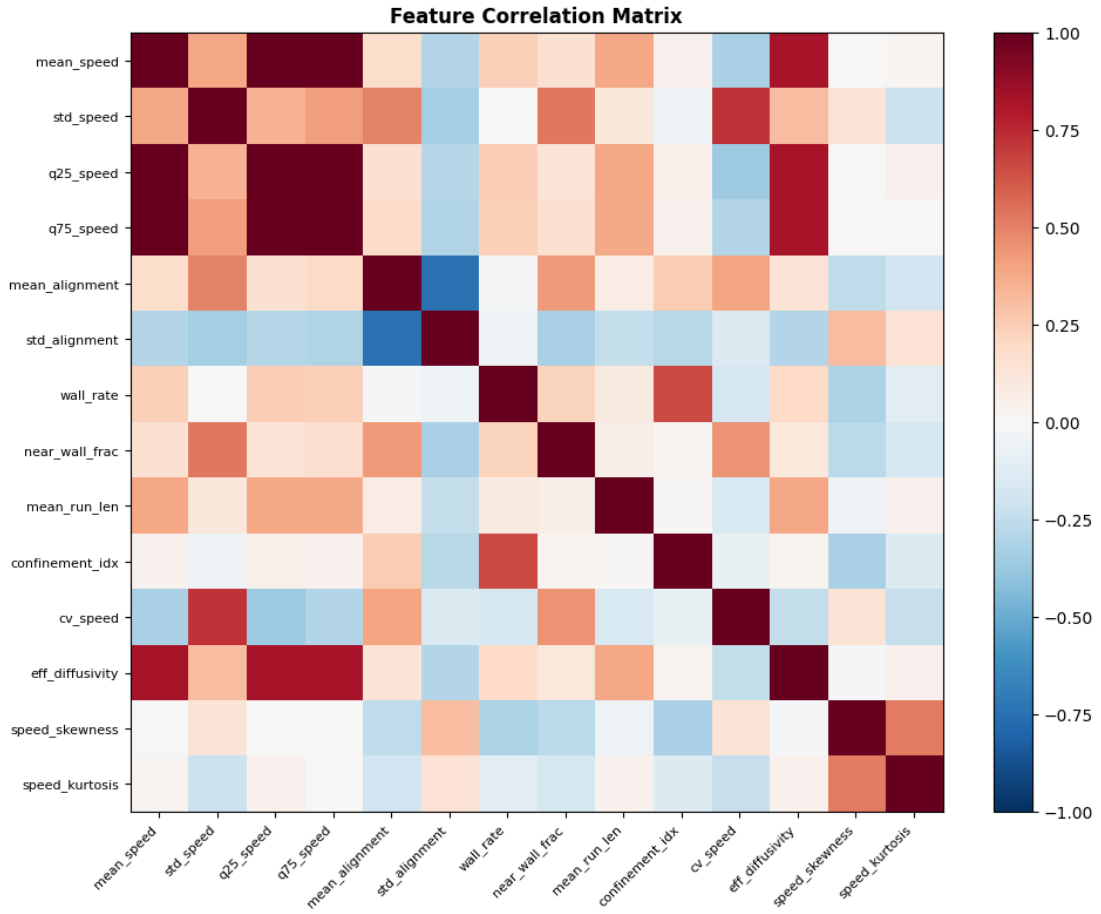


Figure 4.4: Confusion Matrix for Best Performing Model

The confusion matrix in Figure 4.4 shows that Chemotactic and Wall-Guided classes were classified with high precision and recall. The Inhibited class showed limited support due to fewer samples, which affected macro-averaged metrics. However, overall weighted performance remained strong.

4.7 ROC Curve Analysis

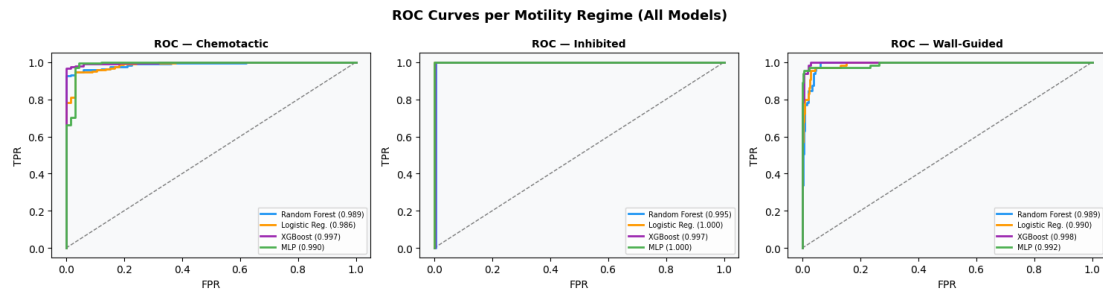


Figure 4.5: ROC Curves for Multi-Class Classification

ROC curves in Figure 4.5 demonstrate excellent separability among classes. The area under curve values exceeding 0.99 confirm strong discriminatory capability of the models.

4.8 Feature Importance Analysis

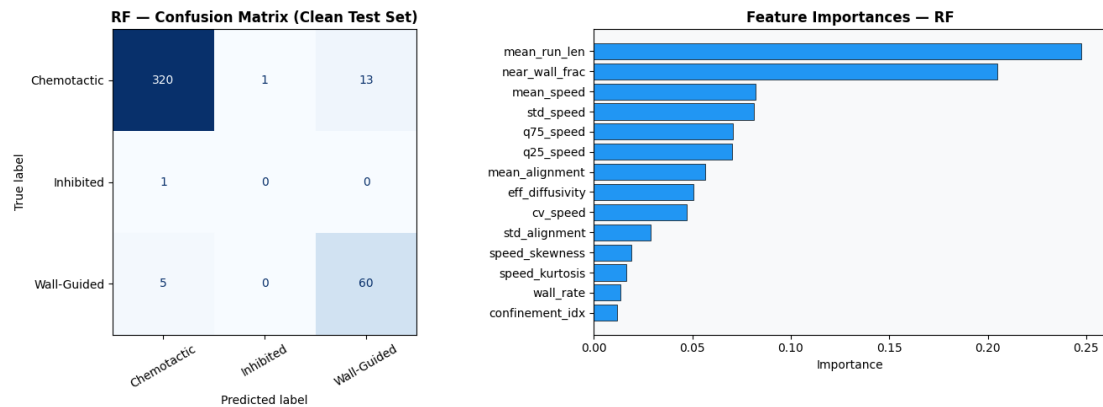


Figure 4.6: Feature Importance Ranking (Random Forest)

Feature importance analysis reveals that mean run length, near-wall fraction, and mean speed are dominant predictors of motility regime. These features align with biological intuition, enhancing interpretability of the model.

4.9 Interactive Dashboard Results

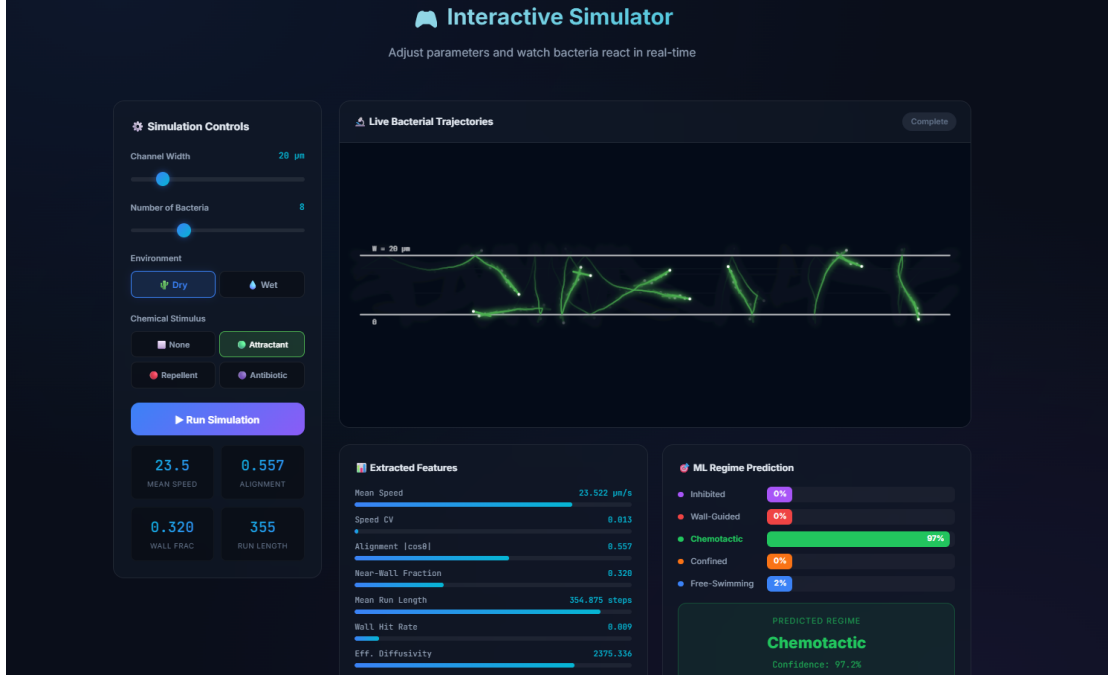


Figure 4.7: Interactive Simulation and Prediction Interface

The developed interactive dashboard allows users to modify channel width, environmental condition, and chemical stimulus in real time. Upon simulation completion, the system extracts features and predicts motility regime with confidence probabilities.

The interface displays trajectory maps, speed histograms, alignment plots, and classification results. This demonstrates practical usability and supports systematic parameter exploration.

4.10 Overall Discussion

The results confirm that the proposed AI-integrated framework successfully captures confinement effects, hydrodynamic interactions, and chemotactic behavior. Simulation outputs align with expected biological patterns, and machine learning models achieve high classification accuracy with minimal overfitting.

The combination of mechanistic modeling and statistical learning provides both interpretability and predictive power. The system demonstrates scalability, repro-

ducibility, and real-time applicability for computational microbiology research.

Overall, the experimental findings validate the effectiveness, robustness, and scientific relevance of the proposed BacMotionAI framework for bacterial motility classification under varying environmental conditions.

Chapter 5

Conclusion

This project presented the design and implementation of a physics-informed and Artificial Intelligence–based framework for analyzing and predicting bacterial motility patterns under varying confinement and chemical conditions. The primary objective was to develop a structured system that combines mechanistic simulation with machine learning in order to classify bacterial trajectories into meaningful motility regimes while preserving biological interpretability.

A comprehensive simulation pipeline was developed to model *E. coli* motility inside a two-dimensional microchannel environment. The system incorporated geometric confinement, dry and wet environmental conditions, hydrodynamic interactions, and chemical stimuli including attractant, repellent, and antibiotic effects. A structured dataset of 2,000 bacterial trajectories was generated using controlled parameter combinations and reproducible random seeds to ensure biological variability and statistical robustness.

From each simulated trajectory, fourteen quantitative motility features were extracted, including speed statistics, wall interaction measures, alignment properties, diffusion behavior, and confinement indices. These features were used to construct a structured dataset suitable for supervised learning. Multiple machine learning models, including Random Forest, Logistic Regression, XGBoost, and Multi-Layer Perceptron (MLP), were trained using a rigorous anti-overfitting pipeline with stratified train-test splitting, feature scaling, SMOTE balancing, and five-fold cross-validation.

The experimental results demonstrate the effectiveness of the proposed frame-

work. The MLP model achieved an accuracy of 98.25

Qualitative trajectory visualization further validated these findings. Narrow channels promoted wall-guided behavior, wide channels enabled exploratory motion, wet conditions introduced hydrodynamic alignment effects, and antibiotic exposure reduced motility as expected. These results confirm that the simulation framework successfully captures biologically meaningful motility transitions.

Furthermore, the development of an interactive web-based dashboard demonstrated the practical applicability of the system. The interface allows users to adjust microchannel width, bacterial count, environmental conditions, and chemical stimuli, while receiving real-time trajectory visualization and regime prediction with confidence scores. This makes the framework suitable for research, educational, and experimental planning applications.

In conclusion, the proposed AI-based bacterial motility analysis system successfully integrates physics-driven simulation and machine learning classification to achieve accurate, interpretable, and scalable motility prediction. The results confirm the reliability, robustness, and research relevance of the framework, making it a promising solution for computational microbiology and intelligent motility analysis.

5.1 Summary

This study focused on the design and development of a Physics-Informed AI Framework for Bacterial Motility Analysis aimed at understanding how confinement, hydrodynamics, and chemical stimuli influence bacterial behavior. Bacterial motility plays a crucial role in infection progression, biofilm formation, chemotaxis, and antibiotic response. Therefore, developing an intelligent and quantitative analysis system is essential for advancing microbiological research.

The study followed a structured methodological approach beginning with simulation environment design. A two-dimensional microchannel of fixed length was constructed with varying channel widths to model strong and weak confinement. Both dry (geometry-only) and wet (hydrodynamic interaction) conditions were implemented to isolate physical effects. Chemical stimuli including attractant,

repellent, and antibiotic conditions were incorporated through parameter modulation of speed and tumbling behavior.

A systematic dataset was generated by varying channel width, environmental state, and chemical condition, producing 2,000 bacterial trajectories. Each trajectory was processed to extract fourteen biologically meaningful features describing speed dynamics, orientation alignment, wall interactions, and effective diffusivity. These features formed the structured input for supervised machine learning models.

The machine learning pipeline included stratified data splitting, feature normalization, SMOTE balancing for class imbalance, and five-fold cross-validation to ensure robust evaluation. Among the evaluated models, the MLP and XGBoost classifiers demonstrated superior performance with high accuracy and ROC-AUC values, confirming strong predictive capability.

In addition to simulation and model development, a user-friendly interactive dashboard was implemented. The system allows real-time parameter adjustment and regime prediction, demonstrating the practical usability of the proposed framework. Overall, this study presents a comprehensive and structured approach for developing an intelligent bacterial motility analysis system using physics-based modeling and machine learning techniques.

5.2 Future Scope

Although the proposed system demonstrates strong performance and promising results, several directions remain for further improvement and expansion:

1. **Extension to Three-Dimensional Simulation and Experimental Validation**

Future work can extend the current two-dimensional simulation to a full three-dimensional motility model to better represent real microfluidic environments. Additionally, validating the framework using experimental microscopy tracking data will strengthen biological relevance and improve model transferability.

2. Integration of Deep Learning–Based Feature Learning and Multi-Species Analysis

Future research may incorporate deep learning approaches such as autoencoders or recurrent neural networks to automatically learn trajectory representations instead of relying solely on handcrafted features. The framework can also be extended to model multiple bacterial species with species-specific motility parameters, enabling broader applicability in microbiology and bioengineering research.

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